1. Explain UEBA

UEBA (User and Entity Behavior Analytics) is a cybersecurity solution that monitors the behavior of users and devices (also called entities) within an organization's network. It uses machine learning and behavioral analysis to understand what is considered "normal" activity and then detects unusual or suspicious behavior that may indicate a security threat.

Unlike traditional security tools that rely mainly on predefined rules or signatures, UEBA identifies activities that deviate from normal patterns. This helps detect insider threats, compromised accounts, and advanced cyberattacks that might otherwise go unnoticed.

Benefits of UEBA

- Detects insider threats.
- Identifies compromised user accounts.
- Detects unusual login and access patterns.
- Reduces false positives by understanding normal behavior.
- Helps SOC analysts investigate suspicious activities more effectively.

2. Differentiate SIEM and UEBA

Although both SIEM and UEBA help improve an organization's security, they serve different purposes.

SIEM	UEBA
Collects and analyzes logs from different devices and applications.	Analyzes user and device behavior to detect anomalies.
Uses predefined rules and correlation to detect threats.	Uses machine learning and behavioral analysis.
Detects known threats based on security events.	Detects unknown or insider threats based on unusual behavior.
Focuses on events and log data.	Focuses on users and entities.
Generates alerts from security logs.	Identifies suspicious behavior even when no rule is triggered.

In many organizations, SIEM and UEBA work together. SIEM collects the security data, while UEBA adds behavioral analysis to improve threat detection.

3. Analyze an Insider Threat Scenario

Scenario

A finance department employee normally works between **9:00 AM and 5:00 PM** and regularly accesses payroll documents.

One night, the employee's account logs in at **2:00 AM** from a different location. Within a short time, the account downloads hundreds of confidential financial files and copies them to a USB drive. The user also attempts to access HR records, which they have never accessed before.

These activities are unusual compared to the employee's normal behavior. A UEBA system detects the abnormal actions and generates an alert for the SOC team.

The SOC analysts investigate the alert and discover that the employee's account was compromised through a phishing attack. The account is immediately disabled, passwords are reset, and the investigation continues to determine if any sensitive information was stolen.

This example shows how UEBA can detect suspicious behavior even when the attacker is using a legitimate user account.

4. List Abnormal Behavior Indicators

Some common indicators of abnormal behavior that UEBA can detect include:

- Login attempts at unusual times.
- Logins from unfamiliar locations or countries.
- Multiple failed login attempts.
- Accessing files or systems that the user does not normally access.
- Downloading or copying a large amount of sensitive data.
- Unusual use of USB storage devices.
- Accessing multiple systems within a short period.
- Sudden privilege escalation or changes to user permissions.
- Large data transfers to external devices or cloud storage.
- Disabling security software or changing security settings.
- Running unfamiliar or unauthorized applications.
- Unusual network activity from a user or device.

These behaviors may indicate an insider threat, a compromised account, or malicious activity and should be investigated promptly by the SOC team.